

Continuous Probability

UChicago Political Science Math Camp

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2026

Continuous Probability

We can expand our understanding of probability to include infinite outcomes

We know that a random variable can model discrete outcomes: i.e., flipping a coin, rolling a dice

What if we took a dice with one face for every real number $\mathbb{R} \in [0, 1]$?

PDFs

A PDF extends a PMF to include a continuous domain, such that

$$f(x) = \Pr[X = x]$$

Where f returns a real number for every input of x . A PDF returns the probability density of a certain outcome of a random variable.

Probability Axioms

Recall our probability axioms:

1. $\Pr[a] \geq 0 \forall a \in A$
2. $\sum_{a \in A} \Pr[a] = 1$
3. For disjoint outcomes, $\Pr[\bigcup_{i=1}^k a_i] = \sum_{i=1}^k \Pr[a_i]$

How might we apply these to continuous probability distributions?

CDFs

Say for some PDF $f(X)$, we want to know the likelihood that x is larger than some X . We can use the **cumulative distribution function**:

$$F(x) = \Pr[X \leq x] = \int_{-\infty}^x f(s)ds$$

We use FTC to obtain this: $\int_a^b f(x)dx = F(b) - F(a)$, for $a = -\infty$, and some arbitrary s

Expanded CDFs

This FTOC definition allows us to specify the probability of an RV falling within a given range. For example,

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We can then use our **disjoint probability axiom** to combine multiple domains:

$$\Pr[x \in [a, b] \text{ OR } [c, d]] = \int_a^b f(x)dx + \int_c^d f(x)dx$$

CDF Foundations

What do our first two axioms imply about continuous CDFs?

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- ▶ Weakly increasing: $F(x) \geq F(x') \forall x > x'$
- ▶ Comprehensive: $F(\infty) = 1$

Joint Distributions

Joint distributions are an easy extension from discrete probability:

$$p(x, y) = P(X = x, Y = y)$$

$$\implies f(x, y); P((X, Y) \in A) = \int \int_A f(x, y) dx dy$$

Probability represented by volume under the joint PDF

Marginal Distributions

A **marginal distribution**, denoted $f_X(x)$, reflects the distribution of one variable, holding the other variable(s) constant.

Then, we can represent by integrating **w.r.t. the other variable!**

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Conditional Probability

To evaluate the **conditional probability** of a continuous RV, we hold the other variable constant:

$$\Pr[y|x] = f_{y|x}(y|x) = \frac{f(x, y)}{f_X(x)}$$

Uniform Distribution

The uniform distribution takes a slightly different form for continuous probability:

$$f(x) = \frac{1}{b - a}$$

for uniformity over the domain $[a, b]$.

Expanding Expectation

Recall that for a discrete RV, $E[X] = \sum_{x \in X} \Pr[x] \cdot x$

How can we expand this to an infinite cardinality of x ?

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$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

Unfortunately, you can't isolate out $\int_{\mathbb{R}} f(x)dx = 1$ here

An Integration Corollary:

The sum rule lends that

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$$

Suppose $f(x)$ is only defined on $[a, b]$ and therefore takes a value of 0 for all other inputs. Then,

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c 0dx$$

$$\implies \int_a^b f(x)dx$$

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Much of probability calls for infinite integrals, but we can often isolate a well-defined domain.

Median

The median defines a value such that half of the mass is above that threshold, and half is below.

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$$\text{Median}(X) \equiv x \text{ s.t. } \int^x f(X) dX = \frac{1}{2}$$